TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Quantum "Ghost" spectroscopy: IR spectroscopy without IR detectors			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (X) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical - Electronics		*Electrical - Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Optical Measurement & Control Devices/Light Sources*	*Measuring Instruments*	*Optical Measuring Instruments*
Concept of Proposed Technology	The proposed technology is a quantum-enhanced "ghost" spectroscopy system that enables infrared (IR) spectroscopic sensing *without direct infrared detection*, bypassing a major technological limitation. The underlying quantum interference effect can be made robust against loss, allowing practical operation outside laboratory conditions, and achieving performance and functionality that are fundamentally inaccessible to traditional classical optical methods. As an added benefit, because they do not require cryogenics, they can be implemented in low Size, Weight, and Power (SWaP) packages; e.g., we have implemented similar quantum light sources for use on drones, and in a small platform on the International Space Station.			
Rationale for Joint R&D Project	While we have demonstrated compelling proof-of-principle results showing genuine quantum advantage in optical sensing and spectroscopy, translating these effects into practical, deployable technologies requires substantial system-level engineering beyond the scope of our laboratory. Real-world utility demands robust, scalable, and field-ready implementations of key components, including stable laser sources, low-noise single-photon detectors, integrated waveguide-based entanglement sources, and nonclassical interferometers that maintain performance under realistic operating conditions. This joint R&D project is essential because these challenges lie at the intersection of quantum optics,			

	photonic engineering, device fabrication, and systems integration. Independently optimizing all these elements while also addressing reliability, packaging, and manufacturability would be challenging for a single group. Close collaboration enables early co-design of optical, electronic, and control subsystems, accelerating progress from laboratory demonstrations to operational prototypes. As academic partner we will contribute established experimental demonstrations, theoretical modeling, and system-level understanding of quantum-interference-based spectroscopy. The industrial partner provides expertise in engineering translation, opto-electronic integration, packaging, and robustness required to advance from a TRL-3 proof of concept to a TRL-6+ prototype. By combining complementary expertise and infrastructure, we reduce technical risk, accelerate development, and establish a clear pathway from scientific validation to deployable quantum-enabled sensing platforms for spectroscopy and remote sensing.
Keywords	1. Quantum-enhanced optical sensing 2. Detector-independent infrared spectroscopy 3. Integrated photonic entanglement sources 4. Low-SWaP optical metrology

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	Our research group is one of the premier quantum optics and optical quantum information efforts worldwide, with over 3 decades of experience in the field, and having performed many seminal experiments on quantum interference and entanglement for sundry applications in sensing, communication and computation. The PI holds 7 patents on various quantum technologies and a strong interest in seeing fundamental quantum phenomena put to practical use. Apropos this proposal, we performed two of the first experiments demonstrating the key underlying phenomena: observing quantum interference in the creation of pairs of entangled photons [1], and realizing quantum interrogation (also known as “interaction-free measurement”) [2], where the presence of a (nontransmitting) object maybe detected with no photons hitting the object, by virtue of the hindrance the object makes to a destructive interference effect [3, 4]. Combined with a source of “energy-entangled” photon pairs - each photon of which has a wide energy spectrum, but the sum of the two energies is very tightly constrained – these interference phenomena allow for “ghost spectroscopy”, in which a substance’s removal of some particular spectral lines in the infrared spectrum causes the appearance of correlated spectral lines in the visible spectrum. In this way IR spectroscopic measurements can be made without detectors that can see in the infrared, or even a tunable IR laser. Encoding the sensing signal in interference contrast rather than direct infrared intensity provides an intrinsic advantage in low-signal and loss-limited environments. The system sensitivity can also be greatly enhanced (by 100-1000) by incorporating an ultra-low loss multipass optical cell. By directing the photons to bounce repeatedly (e.g., 500x) through the same region, one gains a direct improvement, i.e., the effective absorption coefficient is increased by the same factor. We have considerable experience with this technology, having used it to create an ultra-broad bandwidth quantum memory [5]. Moreover, we have devised a method, based on the “quantum eraser” phenomena we first demonstrated in 1992 [6], and subsequently applied to the same sort of pair interference that underlies ghost spectroscopy [1], that allows one to retain high interference visibility (and consequently high spectroscopic sensitivity) even in the presence of substantial loss in the signal returned by the substance under test. Together, these results establish a solid experimental foundation for detector-independent infrared spectroscopy based on quantum interference. Our group has already completed a tabletop experimental verification of the interference process, with results closely matching our theoretical model. Current in-house capabilities: (i) Creation and characterization of nondegenerate energy-entangled photon pairs, with one photon near the visible and the other in the (near) infrared (currently around 1550nm, chosen for ease of use in our proof-of-principle demonstration. We have designs for broadband entangled photon sources creating pairs more suitable for actual mid-IR spectroscopy, with one photon spanning the range 2-5 μm. (ii) We also possess the necessary pump laser technology & single-photon detectors to characterize the entangled photons. (iii) We also have developed a system-level table-top interferometric ghost-spectroscopy setup (Fig. 1) – this laboratory-scale experiment demonstrated spectral reconstruction of a narrowband infrared filter via visible-wavelength detection, without the use of cryogenically cooled infrared detectors (Fig.2). In addition to pump laser & source technology, this includes various alignment and necessary polarization optics, as well as stabilization methods (passive & active). (iv) Relevant to potential future deployment of this technology, we also have an array of systems designed to carry out quantum communication and sensing protocols on mobile platforms like cars & drones. We have used these to demonstrate first-in-the-world drone-to-drone quantum secure communication [7] and even quantum entanglement on the International Space Station [8]. Overall, the R&D objectives are to bridge the gap between a TRL3 laboratory-scale quantum spectroscopy			

demonstration and a deployable sensing prototype (TRL6), establishing a foundation for practical quantum-enhanced and quantum-inspired optical sensing systems. More specifically, we will advance the existing experimental capabilities toward a compact, system-level sensing platform. Planned developments include improving interferometric stability and visibility across broader spectral bandwidths, extending operation toward application-relevant infrared wavelength ranges enabled by the broadband photon-pair source designs, and integrating optical, electronic, and control subsystems into a more robust and portable architecture.

In addition, the project will explore future pathways for mobile and remote sensing implementations, leveraging the inherently low size, weight, and power (SWaP) requirements for visible-wavelength detection. Conceptual deployment on aerial and planetary platforms highlights the potential for remote and in-situ sensing applications. This further enables future extensions to hyperspectral imaging-based sensing modalities, in which spatially resolved spectral information can be obtained without direct infrared detector arrays.

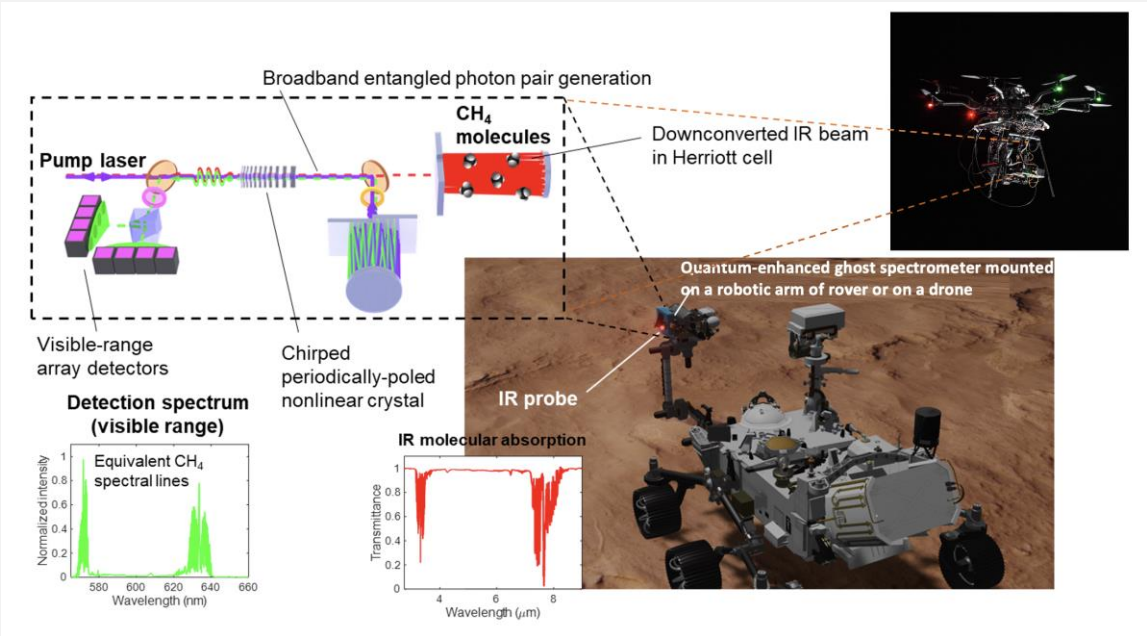

Fig.1: Quantum-enhanced ghost spectroscopy from laboratory demonstration to conceptual mobile platforms. A table-top interferometric setup enables indirect infrared spectroscopy via visible-wavelength detection, while conceptual aerial and planetary deployments illustrate future low-SWaP sensing applications.

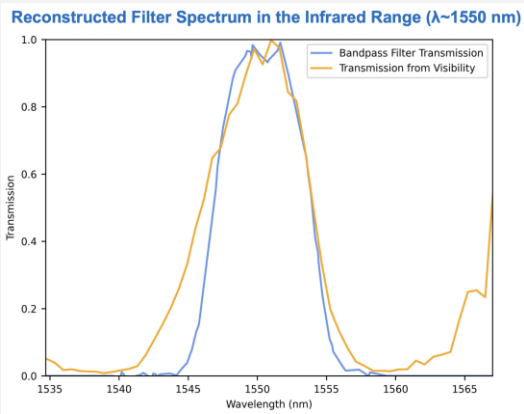

Fig.2: Near-Infrared entanglement spectroscopy: target spectral reconstruction. Comparison between the transmission spectrum of the bandpass filter as measured directly (blue) with a tunable near-IR laser and inferred using the measured visible *photons in our ghost spectroscopy setup (orange).*

R&D Target Specifications

The R&D project targets the development of a compact, quantum-enhanced ghost spectroscopy system that extends the demonstrated laboratory capabilities toward application-relevant infrared sensing, while incorporating robustness, low-SWaP operation, and compatibility with future mobile and imaging-based platforms.

Target Performance Specifications:
- Spectroscopic principle: Quantum-enhanced ghost spectroscopy using interferometric readout
- Detection wavelength: Visible range (room-temperature detectors)
- Infrared sensing range: Near-IR (demonstrated), extending toward mid-IR (2–5 μm)
- Detector requirements: No cryogenic infrared detectors
- System footprint: Initially, rack-mountable prototype, with an eventual goal of a fully integrated robust and portable module.
- Scalability: Compatible with future hyperspectral imaging extensions

Parameter	*Phase-I*	*Phase-II*
SPDC IR range	*1540 - 1565 nm*	*2 - 5 μm*
SPDC VIS range	*802 - 812 nm*	*595 – 725 nm*
IR resolution	*2 nm*	*5 nm*
VIS resolution	*0.16 nm*	*0.1 nm*
Size	50,000 cm^3	< 5000 cm^3

R&D Plan

In conjunction with the Korean SME, the Phase-I effort will fully design the system engineering development plan, as well as identify the critical evaluation methods and metrics. Throughout Phase I, system design choices will be evaluated in the context of low size, weight, and power requirements enabled by visible-wavelength detection.

The Phase-II R&D program will advance the existing quantum-enhanced ghost spectroscopy demonstration toward a compact, system-level sensing prototype suitable for future deployment. The research approach emphasizes experimental validation, robustness, and scalability rather than exploratory physics.

The work will focus on stabilizing and optimizing interferometric performance, extending indirect infrared sensing capability using broadband photon-pair sources, incorporating a low-loss multi-pass optical cell to improve sensitivity, integrating optical and electronic subsystems into a compact architecture, and validating performance against quantitative targets.

By the end of Phase II, the project will deliver a validated system-level prototype and a clear technical roadmap toward mobile, remote, and imaging-based quantum-enhanced sensing platforms. In particular, we will work to deploy the functional prototype on one of our mobile platforms (e.g., one of our Alta-8 drones).

Timeline: Key tasks and milestones:

Phase I	*Key tasks and milestones*
Period	
Months 1–2	*Initial meetings with SME to discuss basic goals, capabilities and principles of operation for ghost spectroscopy*
Months 3–4	*Identification of critical metrics and design "pain points". Preliminary design of system architecture compatible with eventual low-SWaP implementation*
Months 5–6	*Co-design of final prototype, including 1-2 risk-reduction alternatives. Creation of Phase-II proposal.*

Phase II	*Key tasks and milestones*
Period	
Q1 & Q2	*Acquisition of required new hardware, e.g., broadband SPDC source, tunable filters, stabilization hardware.*
Q3	*Testing and verification of components*
Q4 & Q5	*Assembly and optimization of ghost spectroscopy architecture*
Q6	*System evaluation and design of version upgrade*
Q7	*Acquisition of upgrade materials and components*
Q8 & Q9	*Installation of system version 2 upgrades. Performance evaluation of prototype.*
Q10	*Proof-of-concept deployment on mobile platform. Planning for future development to higher (>6) TRL.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (UIUC) will lead the core scientific and technical development of the quantum-enhanced ghost spectroscopy architecture. Responsibilities include system-level design of the interferometric measurement scheme, optimization of quantum interference and quantum eraser–based configurations, and experimental validation of infrared spectral reconstruction using visible-wavelength detection. We will provide theoretical modeling, performance analysis, and experimental data to guide system optimization. We will also support integration strategies, help define performance benchmarks, and contribute to joint evaluation of scalability toward compact and portable implementations. Finally, we will attempt a first deployment on one of our mobile platforms. Throughout, we expect to co-design with the SME, to efficiently identify/create components that will enable optimized ghost spectroscopy on a low SWaP architecture.
Expected R&R of Korean SME Partner	Our current ghost spectroscopy implementation had the sole goal of validating the basic principle of operation, which it did. However, it occupies much of an optical table, and is not sufficiently robust for long-term operation. We rely on the SME to help design and create a prototype that will have higher sensitivity (e.g., incorporating a multi-pass cell and the addition stabilization that requires), greater resilience to background (e.g., by incorporating single-mode fiber collection), much broader bandwidth in the IR (and optimized spectrographic analysis of the visible photons), and much lower form factor, incorporating integrated or microbench optics where possible. Finally, the SME will assist in developing an advanced control and analysis system (to extract the spectroscopic signature from the data) and a user-friendly interface.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcome of this joint R&D project is the demonstration of a compact, quantum-enhanced ghost spectroscopy system that extends validated laboratory capabilities toward application-relevant infrared sensing. Technically, the project will establish a system-level architecture enabling infrared spectroscopy without cryogenic detectors, while maintaining robustness against loss and low signal levels. We will realize and evaluate a pilot system (TRL6) that we can then test on a mobile platform (which necessarily requires a low SWaP solution). Finally, the results will define clear performance benchmarks and design pathways for potential future hyperspectral imaging-based implementations. From an impact perspective, the project lays the groundwork for next-generation optical sensing technologies that reduce system complexity and power requirements compared to conventional infrared spectrometers. The outcomes support future commercialization opportunities in sensing, monitoring, and inspection applications, while strengthening international collaboration and advancing industrial readiness for quantum-enabled measurement technologies.

6. References

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